Lab Assignment Week 09

CSC 3320 – System-level Programming

Thursday March 12th, 2026

Introduction

Welcome to the 9th programming lab of CSC 3320! Today, we will be covering the following topics:

1. Structs in C
2. Pointers to structs
3. Passing structs to functions using pointers
4. Arrays of structs

Structs allow programmers to group related data together into a single data type. When structs are combined with pointers, they allow functions to efficiently access and modify complex data structures. In this lab, you will implement a program that manages student records using structs and pointers.

Lab Policies

- Attendance is mandatory.
- Labs must be completed **individually.**
- TAs are here to help you. Ask them for help!
- Lab assignments are due at midnight on the day of your lab.

Deliverables:

1. The C Code for your program. (.c file).
2. A screenshot of the output in the Terminal.

If you have any questions, please do not hesitate to ask your TA.

Program: Student Records using Structs and Pointers

Write a C program that manages a small set of student records using **structs and pointers**.
First, define a struct called Student that contains the following fields:

- name (character array of size 50)
- id (integer)
- gpa (float)

Your program should create an **array of three students** and use **pointers to access and modify the data**.
You will also implement several functions that operate on these student records.

Required Functions

Your program must implement the following functions.

Function 1

This function prints the information of a student.

```
void printStudent(const struct Student *s);
```

The function receives a pointer to a Student struct and prints the student's name, ID, and GPA.

Function 2

This function updates the GPA of a student.

```
void updateGPA(struct Student *s, float newGPA);
```

The function should modify the student's GPA using the pointer parameter.

Function 3

This function finds the student with the highest GPA.

```
struct Student* findTopStudent(struct Student arr[], int size);
```

The function should return a pointer to the student with the highest GPA in the array.

Program Requirements

Your program should perform the following steps:

1. Create an array of three Student structs.
2. Initialize the student data in the program.
3. Use a pointer to traverse the array of students and print their information.
4. Call updateGPA() to modify one student's GPA.
5. Use findTopStudent() to determine which student has the highest GPA.
6. Print the top student's information.

You must use the arrow operator (->) when accessing struct members through pointers.

Example Output

```
All Students:

Name: Alice Johnson
ID: 101
GPA: 3.50

Name: Bob Smith
ID: 102
GPA: 3.70

Name: Carol Davis
ID: 103
GPA: 3.95

Updating Alice's GPA...

Name: Alice Johnson
ID: 101
GPA: 3.90

Top Student:

Name: Carol Davis
ID: 103
GPA: 3.95
```

Deliverables

For today's lab, you will need to upload the C program code for your bubble sort program and its output in the terminal on iCollege. Please name your C code and screenshot as follows:

- C Files
  - lastname_firstname_filename.c
  - For example: **turing_alan_student_records.c**
- Screenshots
  - lastname_firstname_filename.png
  - For example: **turing_alan_ student_records.png**